

MachBoost: Speculative Decoding from Local Context for On-Device LLMs

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Abstract

Autoregressive decoding remains serial even when a model fits comfortably on a laptop. MachBoost tests a narrow way around that constraint: use tokens from explicit local context as speculative candidates, verify a block with the target model, and return to the runtime’s native decoder when no candidate exists. The current implementation combines context-only n-gram lookup, block verification, in-place MLX KV-cache rewind, verifier continuation tokens, native-compatible prompt prefill, and `mlx-lm` streaming. It also provides a resident local server so repeated chat and completion requests reuse one loaded model. We compare the decoding method with native cached generation rather than a synchronous Python token loop. On an Apple M1 Max, using `mlx-community/Qwen2.5-3B-Instruct-4bit`, the default profile reaches a $1.96\times$ median paired speedup on a 64-token code-continuation fixture over five fresh-nonce repeats. An experimental one-token re-entry profile reaches $1.58\times$ on copied retrieval answers. All 30 rows across the two profiles match their paired native greedy outputs; open-ended controls stay near the native path. Separately, five warm resident 64-token requests take a 0.657-second median wall time, and short chats reach first text in a 0.298-second median. The results fall short of a universal $2\times$ accelerator. They identify a conditional operating region and separate algorithmic gains from model residency and serving overhead.

1 Introduction

Running LLMs on a laptop has become realistic, but it has not made autoregressive decoding disappear. A small model on Apple Silicon may fit in memory and still feel slow because the loop is serial: produce a token, update the prefix, run the model again. That bottleneck is especially visible in developer workflows, where the model often sits inside an editor, command runner, notebook, or local assistant.

Speculative decoding offers a way around part of this loop. A cheap source guesses several future tokens; the target model checks the guess; accepted tokens are emitted without paying for one full target step each. Prior work usually gets the draft from another model, extra heads, or a reference sequence. MachBoost takes the reference idea and applies it to the local machine. If the user is asking about a repo, config, note, policy, or retrieved passage, the likely continuation may already be present in that local text.

The scope is deliberately modest. MachBoost does not try to accelerate every prompt. It tries to accelerate grounded prompts without changing the target model’s greedy output. If the local text is unhelpful, the method can be slower; the implementation exposes that fact through measurement and falls back to the backend’s native path.

This paper contributes the following:

- a context-only drafter and initial-candidate gate that avoid treating the prompt wrapper itself as external evidence;
- a cache-aware MLX verifier that fuses continuation and draft scoring, rewinds rejected cache entries in place, and resumes native decoding when useful context ends;
- a paired benchmark against native `mlx-lm` generation with raw rows, run-order alternation, environment provenance, and exact token comparisons;
- a resident local server with warm model lifecycle, streaming Ollama-compatible and OpenAI-compatible endpoints, and raw completion support for editor integrations;
- negative experiments with a small draft model, an MTP draft path, DFlash, an Apple Neural Engine draft, and lossless weight compression.

2 Background and Related Work

Speculative decoding was introduced as a way to accelerate autoregressive transformer inference without changing outputs by drafting candidate tokens and verifying them in parallel with the target model [6]. Speculative sampling independently developed a related draft-and-verify method for sampled decoding and reported 2–2.5 $\times$ speedups in a distributed setting [4]. Subsequent work explored architecture-level or self-drafting variants. Medusa adds multiple decoding heads to predict future tokens and verifies candidate trees [3].

MachBoost is closest to methods that exploit references or prompt overlap rather than an auxiliary draft model. LLMA observes that outputs in retrieval and reference-grounded scenarios often share spans with the input reference, then verifies copied spans to preserve greedy output [13]. Prompt Lookup Decoding replaces the draft model with n-gram matching over the prompt [8]; PLD+ extends that idea by using language-model artifacts such as attention or hidden states to rank candidate spans [9]. Hugging Face Transformers exposes prompt lookup through `prompt_lookup_num_tokens` in assisted generation [5]. MachBoost differs mainly in packaging and operating assumptions: it treats local files and retrieved snippets as ordinary inputs to the accelerator, keeps benchmark data machine-readable, and makes low-overlap prompts a measured fallback case rather than an error.

Results above 2 $\times$ elsewhere generally require more than process tuning. ReDrafter trains a recurrent draft conditioned on target hidden state and reports up to 2.3 $\times$ on Apple Silicon [14]. QuantSpec changes the draft’s weight and KV-cache representation and reports speedups up to about 2.5 $\times$ in long-context inference [10]. Mirror-SD maps complementary draft and target work across GPU and NPU devices and reports 2.8–5.8 $\times$ on server-scale models [2]. At the runtime layer, BaseRT’s native Metal kernels report up to 1.35 $\times$ over MLX decode rather than a universal 2 $\times$ [7]. These are different operating points: trained model changes, heterogeneous execution, or low-level kernels. MachBoost’s current context path needs none of them, but only applies when the requested continuation overlaps local text.

The same distinction appears outside text generation. Diffusion workloads can reuse intermediate transformations across denoising steps; EasyCache reports 2.1–3.3 $\times$ for several video models [15]. That mechanism does not transfer directly to autoregressive text decoding. It does, however, support a broader product design in which one package selects a backend-specific accelerator rather than pretending that a single algorithm fits text, image, audio, and video models.

The implementation targets common local inference stacks. Hugging Face Transformers provides the broad model API surface for Python inference [11]. MLX is an Apple Silicon-oriented

array framework and runtime used here for Mac-native evaluation [1]. Our measurements use a Qwen-family model converted for MLX; Qwen3 is the relevant public model-family report [12].

3 Problem Setting

Let M be a target autoregressive model and let x be the prompt token sequence. Greedy decoding emits

$$y_t = \arg \max_v M(v \mid x, y_{<t})$$

for each generation step t . MachBoost receives a corpus C derived from prompt text, retrieved context, local files, or user-provided strings. The goal is to reduce wall-clock latency while returning the same sequence y that greedy decoding with M would return.

This paper stays with greedy decoding. Sampling-compatible speculative decoding is possible in principle, but greedy argmax verification gives a simple correctness check: the accelerated sequence must match the serial baseline token-for-token.

4 Method

The implementation has a drafter, a verifier, and an escape hatch to native generation. The escape hatch is important: speculative decoding is a conditional optimization, not a replacement decoder.

4.1 Drafting from Nearby Text

The drafter indexes token n-grams from a corpus C . At generation time it looks at the current prefix $p = x \parallel \hat{y}$, searches for suffixes of p that occur in C , and proposes the tokens that followed the best matching span. Longer suffix matches are preferred, with draft length as a secondary signal. Only explicit context is indexed in the reported adaptive mode. An earlier implementation indexed the concatenation of prompt and context; because benchmark prompts already displayed the source passage, that design could draft from the prompt wrapper and blur the boundary between available context and generated history.

Before entering the verifier loop, MachBoost asks whether the generation boundary has a context candidate. If not, the default profile calls native `mlx-lm` generation immediately. An opt-in re-entry profile generates a bounded number of target tokens, checks the context index again, and either enters block verification or resumes native generation from the live cache. Re-entry can recover copied answers whose first token begins a span in retrieved text. It also changes the cache trajectory, so version 0.2.0 disables it by default.

4.2 Verification Against the Target Model

Given a prefix p and draft $d = (d_1, \dots, d_k)$, the verifier checks how many draft tokens equal the target model’s greedy continuation. A verifier-capable runtime scores a block in one target call and compares target argmax tokens at the relevant positions. If the first a draft tokens are accepted, they are committed. On a mismatch, the target’s residual token is emitted. When the whole block is accepted, the verifier also returns the next target token already present in the logits. The following round scores that pending token together with the next draft, avoiding a separate target call between accepted blocks.

Under deterministic target arithmetic, the intended invariant is:

$$\text{Boost}(M, x, C, T) = \text{Greedy}(M, x, T)$$

for a maximum generation length T . Batched and scalar floating-point paths can still choose different argmax tokens at a near tie, so this is tested rather than assumed. Every benchmark row records an `output_match` flag from the native and accelerated token IDs.

4.3 Keeping the Cache Path Stable

Verification writes candidate entries to the MLX prompt cache. Accepted entries remain useful; rejected entries must not survive. MachBoost trims the rejected suffix in place when the cache type permits it. If safe trimming is unavailable, the service reconstructs the accepted prefix. Prompt initialization follows `mlx_lm`'s split prefill trajectory, processing the final prompt token separately. Once no further context candidate exists, the service hands the live verifier cache to `mlx_lm.generate_step`; the remaining response therefore uses the runtime's optimized asynchronous decoder instead of a Python-level `argmax/synchronize` loop.

4.4 Deciding When to Use the Drafter

Drafting from local text is sometimes the wrong choice. Open-ended prompts often accept only a few draft tokens, so verification overhead dominates. MachBoost therefore exposes a benchmark step. For a prompt family, it measures:

- token match between baseline and boosted output,
- wall-clock speedup,
- accepted draft tokens divided by generated tokens,
- target-call or forward-call reduction.

The high-level API can also calibrate an accelerator across representative prompts:

```
from machboost import Accelerator, GatePolicy

boost = Accelerator.from_mlx(model, context_paths=["./docs", "./src"])
calibration = boost.calibrate(
    prompts,
    max_tokens=32,
    gate_policy=GatePolicy(min_speedup=1.05,
                           min_acceptance_rate=0.10),
)
text, stats = boost.generate(prompt, max_tokens=128)
```

If the measured workload fails the policy, `generate` uses the native backend path. Calibration and the per-request initial-candidate gate serve different purposes: calibration enables a workload family, while the gate prevents verifier overhead on an individual request with no draft at its boundary.

5 Implementation

MachBoost is implemented as a small Python package with no required runtime dependencies. Optional extras install backend-specific dependencies:

- `machboost[mlx]` for MLX and `mlx-lm`,
- `machboost[hf]` for PyTorch and Transformers,
- Ollama HTTP support for benchmarking and capability detection.

The core package defines a `StepService` protocol with `next_token(prefix)` and an optional `verify(prefix, candidate)` hook. A service with only `next_token` can use the wrapper but cannot skip target work. A verifier can accept several draft tokens per target call. The high-level `Accelerator` loads MLX or Hugging Face models, reads strings, files, or directories as context, and exposes `benchmark`, `calibrate`, and `generate`.

Version 0.2.0 adds a resident HTTP process. It owns loaded accelerators, serializes requests per model, retains each model indefinitely by default, and supports finite idle lifetimes or explicit unloading. The CLI can start this process automatically, preload a short model alias such as `qwen2.5:3b`, and issue chat or raw completion requests without reloading weights. The server exposes `/api/chat`, `/api/generate`, model lifecycle endpoints, and OpenAI-compatible chat and completion endpoints. Native MLX streaming forwards `mlx-lm`'s already-detokenized text segments; verified multi-token spans use a stateful streaming detokenizer. This avoids the earlier cost of decoding the entire token prefix once per emitted token.

The Ollama HTTP adapter remains a wrapper: Ollama's public generation API does not expose the target token IDs, logits, mutable KV cache, and block-verification hook this implementation needs. Native integration would require a runner-level change rather than an HTTP option.

6 Evaluation

6.1 Machine and Model

Measurements were collected on an Apple M1 Max with 10 CPU cores and 32 GB unified memory, running macOS 26.5.2. The environment used Python 3.13.3, MLX 0.31.2, and `mlx-lm` 0.31.3. The paired decoding artifacts used MachBoost 0.1.4; the resident serving artifact used version 0.2.0. No thermal or performance warning was reported immediately before or after the paired run. The model was `mlx-community/Qwen2.5-3B-Instruct-4bit`.

The benchmark requests 64 greedy tokens for three fixture families: literal Python continuation, retrieval-grounded answering, and open-ended writing. Each family is repeated five times with a fresh nonce under both default and re-entry profiles. Baseline and accelerated order alternate. Wall time includes prompt processing. The baseline calls `mlx-lm`'s native streaming generator; it is not MachBoost's `next_token` loop. The artifacts `results/mlx_native_default_qwen25_3b_20260713.json` and `results/mlx_native_reentry_qwen25_3b_20260713.json` contain prompts, token previews, raw times, package versions, hardware data, and each run's order.

6.2 Native-Baseline Results

Profile/path	Match	Median	Base tok/s	MB tok/s	Accepted
Default/code	100%	1.96×	87.46	167.28	51
Default/RAG	100%	1.04×	89.98	91.61	0
Default/open	100%	1.00×	99.95	98.27	0
Re-entry/code	100%	1.62×	72.54	121.38	51
Re-entry/RAG	100%	1.58×	88.92	140.09	30
Re-entry/open	100%	1.08×	94.25	101.57	0

Table 1: Five fresh-nonce repeats per profile and fixture. Speedup is paired wall time; throughput and accepted-token columns are per-column medians. “MB” denotes MachBoost.

The default code fixture begins at a boundary that occurs in the supplied source. It accepts a median 51 of 64 output tokens and reduces logical target forwards by 76.6%. Its paired speedups are 2.44, 1.15, 2.08, 1.96, and 1.91×. The 1.96× median is close to the requested 2× threshold but does not clear it, and the range shows how noisy short local measurements remain.

The default retrieval answer has no candidate at its generation boundary and takes native fallback. With one-token re-entry and 1-gram lookup, the first target token anchors a context span; the verifier then accepts a median 30 tokens and reaches 1.58×. Open-ended generation accepts no draft tokens. Its apparent 1.08× ratio is timing noise rather than an algorithmic gain.

All 30 reported rows match their paired native token sequences. A separate 15-row exploratory run with a three-token re-entry probe had one RAG mismatch after an accepted span. This is why re-entry remains opt-in. For a user-facing system, the useful statistics are coverage, conditional speedup, and equality rate rather than one aggregate median.

6.3 Resident Serving Results

The resident benchmark preloads the same 3B model, then sends five forced 64-token completions and five short streaming chats. It measures wall time at the server or Python streaming client, including prompt processing and detokenization. Table 2 summarizes the artifact `results/resident_qwen25_3b_20260713.json`.

Workload	Rows	Median TTFT	Median total	End-to-end tok/s
Forced 64-token completion	5	–	0.657 s	97.47
Short streaming chat	5	0.298 s	0.358 s	–

Table 2: Resident-server latency after preloading the model. Chat outputs contain nine tokens in each recorded row; TTFT is measured when the client receives the first non-empty text segment.

The first forced completion takes 0.973 seconds because the Metal execution path still needs initialization after weight loading; the following four take 0.650–0.663 seconds. These rows accept no context draft tokens. The measurement therefore demonstrates load amortization, corrected streaming, and native fallback rather than a speculative-decoding gain. A same-day Ollama probe records a 0.883-second median for the same forced-token shape, a 1.35× wall-time ratio relative to the 0.657-second resident run. The two runtimes use different 4-bit model formats and the requests were not interleaved, so this is a backend-suitability observation rather than exact-weight evidence.

6.4 Why Earlier Ratios Were Rejected

Earlier repository artifacts reported 3–8 $\times$ on MLX. Their baseline disabled the prompt cache or performed a synchronized target call for each token. That path ran near 10 tokens/s and was not representative of native `mlx-lm`, which uses cached asynchronous generation. Those artifacts remain available as implementation diagnostics, but they cannot support a production speed claim. Replacing that baseline was also what exposed a regression in MachBoost’s serial tail; switching the tail to native generation restored parity for non-overlap prompts.

A one-repeat Hugging Face artifact likewise found MachBoost context lookup competitive with Transformers prompt lookup, but it predates the current paired harness. It is useful for designing a larger comparison, not for the headline result.

6.5 Exploratory Alternatives

We tested several ways to extend acceleration beyond literal context overlap. Table 3 summarizes the local probes. They were not all run under one benchmark protocol, so the numbers are diagnostic rather than comparative.

Path	Local result	Main constraint
DFlash, Qwen3.5 9B	0.66 $\times$ at 128; 0.96 $\times$ at 512	M1 lacks the newer accelerator path; draft emulation costs more than accepted work saves.
Qwen2.5 0.5B draft for 3B target	0.38–0.78 $\times$	Acceptance of roughly 36–58% did not repay serial draft generation.
Ollama embedded MTP	12.7–24.3 tok/s versus roughly 40 tok/s baseline	The tested draft depths added overhead on this model and machine.
ANE 0.5B draft	51.1 decode tok/s	A one-token ANE draft cannot feed a 3B MLX target near 102 tok/s economically; multi-token heads are needed.
Lossless Q4 compression	1.08–1.15 $\times$ ideal entropy bound	The quantized weights are already close to high entropy; decoder cost would reduce the gain further.

Table 3: Exploratory paths that did not beat native generation locally. Each result is specific to the tested model, implementation, and M1 Max.

The failures share a simple accounting problem. A draft only helps when the time removed from target decoding exceeds draft generation, verification, synchronization, and cache repair. A smaller model is not automatically a cheap enough draft. Moving it to another processor is not sufficient either if it predicts one token at a time. The strongest published results solve this with trained multi-token prediction, high acceptance, or concurrent heterogeneous execution rather than with process priority.

7 Limitations

The native-baseline evidence has 30 headline rows, one model, one quantization, one machine, and controlled 64-token prompts. The resident evidence adds ten requests but does not broaden model or hardware coverage. The accelerated code fixture is intentionally favorable: the desired code is substantially present in context. It demonstrates the mechanism but does not estimate

how often real users encounter an eligible boundary. Longer repository sessions, multiple 3B–9B architectures, prefill-heavy prompts, concurrent clients, and energy measurements remain untested under the corrected harness.

Only greedy decoding is evaluated. Exact token equality under the tested execution paths does not prove equality for sampling, beam search, or every floating-point near tie. Images, audio, and video are not accelerated by the current package. Their models need modality-specific adapters; applying text n-gram verification to them would be a category error. Ollama is also not accelerated through its public HTTP API.

Finally, this work does not introduce prompt lookup itself. The engineering contribution is the adaptive package path, MLX cache integration, native fallback, and reproducible baseline correction. A novelty claim beyond that would need broader comparisons with LLMA, Prompt Lookup Decoding, PLD+, and current trained speculative systems.

8 Next Experiments

There are two plausible tracks. The first keeps models unchanged: add an online eligibility predictor, extend bounded re-entry beyond the current one-token experiment, and add native verifier hooks for llama.cpp/Ollama. The resident server makes this testable under long sessions; evaluation should report request coverage, cold and warm time to first token, decode throughput, peak memory, concurrency, energy, and exact outputs against each backend’s own optimized generator.

The second track accepts model-specific preparation. A multi-token drafter distilled from the target could run on the Apple Neural Engine while target verification runs on the GPU, following the heterogeneous lesson of Mirror-SD. Quantized draft weights and KV state could reduce memory traffic in long contexts. Both ideas exceed the scope of a wrapper and should be treated as separate research prototypes with quality and compatibility tests.

For image and video generation, the package can expose the same routing and evidence interface while using activation or transformation caching such as EasyCache. This would make MachBoost a common control plane for specialized accelerators, not a claim that one decoder optimization applies to every model.

9 Conclusion

MachBoost does not yet deliver a stable universal $2\times$ improvement. On a 3B MLX model, literal context-backed code continuation reaches a $1.96\times$ median over five paired runs; one-token re-entry broadens coverage to copied retrieval answers at $1.58\times$. All 30 headline rows match native greedy token sequences, while requests with no accepted draft remain near the native path. A resident server now separates model loading from request latency: warm 64-token requests take a 0.657-second median, and short chats reach first text in 0.298 seconds. The corrected evidence makes the next problem precise: increase acceptance without a slow serial draft or a divergent cache trajectory, then validate the result across models, prompts, sessions, and modalities.

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